HARIPRASATH PERIYASAMY

Principal Engineer · Product & Web Engineering · Applied AI

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Boston, MA, USA · Open to relocation

PROFESSIONAL SUMMARY

Product engineer with 10+ years of full-stack, frontend-heavy work, most of it shipping 0-to-1 products at AWS and Shopify. Launched Amazon Ads' Ads Data Manager and made it 2.6x faster at peak, built the CloudWatch Logs Live Tail front end streaming 10,000+ events/second, and led Shopify's Ads 3P team through a dynamic creative system serving ads in under 150 ms. Now Principal Engineer at Autodesk, where I proposed and built a human-in-the-loop AI agent that finds the root cause of production problems, had it in production through enterprise AI governance within five months of joining, and am now scaling it into a framework for all of Autodesk.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Product & Web: React, Next.js, Redux, React Query, Node.js, GraphQL, REST, WebSockets, Web Workers, Server-Driven UI, Design Systems, Accessibility (WCAG), Mobile & Touch Web

Performance & Scale: Web Vitals, Query Batching & Multi-Level Caching, Streaming Data UIs, Memory Profiling (Chrome DevTools), Bundle Optimization, Telemetry & SLIs, Distributed Tracing

Applied AI: Agent Orchestration (LangGraph), Amazon Bedrock, Model Context Protocol (MCP), Context Engineering, RAG, Knowledge Graphs (Neo4j), LLM Evaluation, Human-in-the-Loop Design, AI Risk & Security Review, Agentic Coding Tools (Claude Code and similar)

Cloud & Quality: AWS (CDK, Lambda, S3, CloudWatch; multi-account, multi-region), Docker, Kubernetes, CI/CD, Jest, Cypress, Playwright, React Testing Library, A/B Experimentation, Staged Rollouts

PROFESSIONAL EXPERIENCE

Principal Engineer | **Autodesk** | Remote (*May 2026 – Present*)

Post-Purchase – Autodesk Account Subscriptions & Billing

- **Proposed and built an AI agent that finds the root cause of production problems** (TypeScript, LangGraph, Amazon Bedrock, MCP, Neo4j), drawing on years of CloudWatch Logs work: it scans full logs for patterns instead of relying on status codes and generic observability signals, grounded in an ontology graph built by an automated harvester. In production within five months of joining, it cut investigation time **from ~30 to ~5 minutes**, saving **~25 engineer hours a month**.
- Designed it for responsible use: retrieval over the ontology graph, read-only access through allow-listed integrations, no automatic remediation, an engineer reviews every action, and an evaluation harness and verification loop check answers are grounded (**~90% correct on first answer**); took it through enterprise AI governance, risk and security review to **AWS production with 1,200+ automated tests**.
- **Now scaling the agent into a framework for all of Autodesk.** Turned my path through AI governance into the playbook the next team uses to ship governed AI (approval in ~4 weeks instead of ~3 months), and started the AI adoption forum that sets AI priorities for three teams, turning 12+ demos into an owned backlog; the teams now share the 10-server MCP tool stack, agent skills and spec-driven, test-first agentic practices I standardized.
- Lead engineer for Autodesk's personalized, AI-driven subscriptions and billing web application, used by **thousands of customers daily**; led front-end delivery of the flagship subscriptions redesign **across a 13-person team** with product and design, cutting billing support tickets ~20%.
- Wrote the **UI readiness framework 16 engineers follow** as their quality bar, cutting UI defects reaching production ~30%, and proposed distributed tracing across services; built shared design-system components now **reused by three sister teams**, so copy and visual changes ship in hours instead of ~1 week.

Senior Engineer | **Shopify** | Remote (*Aug 2025 – May 2026*)

Ads 3P – Real-Time Bidding, Creatives & Merchant Marketing

- **Led the Ads 3P team (8 engineers)** across real-time bidding, merchant dashboards and email/SMS automations, setting its architecture direction and mentoring engineers on code quality and delivery velocity.

- Designed and shipped a dynamic creative optimization system end to end, from UI rendering to the data layer, assembling and serving ad creatives in **under 150 ms** and lifting click-through ~12%.
- **Represented Shopify in architecture reviews with Microsoft's and Instacart's engineering teams** on real-time bidding, leading to ad creatives that load ~40% faster.
- Built high-integrity event tracking across web and native clients, delivering reliable real-time conversion signals to Meta Pixel and Google Tag Manager and **recovering ~15% of previously lost conversions** for merchants.
- Carried on-call for the team's services, driving security hardening and system-health fixes that cut pages ~30%; ran structured code reviews and knowledge-sharing sessions.

Front-End Engineer | **Amazon Web Services (AWS)** | Boston, MA (*Jun 2022 – Aug 2025*)

Amazon Ads – Full-Stack Digital Experiences

- Launched Ads Data Manager **0-to-1**, an enterprise advertiser data onboarding and activation application (React, TypeScript, GraphQL) adopted by 500+ advertisers, with chunked-streaming uploads for files up to 5 GB.
- Architected a GraphQL Backend-for-Frontend layer stitching multiple microservices into one client contract; query batching and multi-level caching (client, CDN) **cut backend calls 60% and average response time from 850 ms to 320 ms** at peak traffic.
- Designed the **server-driven UI architecture Ads Data Manager is built on**, delivering UI configuration from the backend so the team ships interface changes and A/B experiments **without client releases**, from ~2 weeks to same-day.
- Gave the team secure self-service deployments across multiple AWS accounts and regions with AWS CDK, moving releases from weekly to daily.

AWS CloudWatch – Real-Time Monitoring & Observability

- Built the CloudWatch Logs Live Tail front end on WebSockets and Web Workers, **streaming 10,000+ log events/second** while keeping the UI responsive, so customers debug in real time.
- Built CloudWatch's ML-driven log anomaly detection dashboards, turning raw model output into interactive visualizations that **let non-technical users act on anomalies**, cutting time to spot one from hours to minutes.
- Instrumented client-side telemetry (latency, API errors, behavioral metrics) and used it to find and fix ~20 performance regressions and guide product decisions.

Senior Frontend Engineer | **Sumo Logic** | Boston, MA (*Sep 2021 – May 2022*)

- Cut data-analytics dashboard load time 45% (tree shaking, lazy loading, SLI-based monitoring) and memory use 25% by fixing leaks in large data-visualization components; established the Atomic Design-based design system the product suite is built on, speeding new dashboard builds ~30%.

R&D Software Engineer | **Dassault Systèmes** | Boston, MA (*Jan 2019 – Aug 2021*)

- Owned the Web UI template library for xApps (browser-based CAD applications) that every feature team company-wide (20+ teams) built on, with mobile and touch support and load/runtime optimizations (Web Workers, caching, compression).

Programmer Analyst | **Cognizant Technology Solutions** | India (*Jul 2014 – Nov 2016*)

- Built accessible, cross-browser interfaces for digital banking platforms including Lloyds Bank Commercial Banking Online, and a test-automation framework that cut manual testing 25%.

EDUCATION & COMMUNITY

Master of Science in Informatics, Northeastern University, Boston, MA (Jan 2017 – Dec 2018)

ADPList Mentor – Recognized as a Top 10 UX Engineering Mentor